HE2001 Microeconomics II

Tutorial 4

Academic Year 2025/2026, Semester 2

Quantitative Research Society @NTU

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Overview

This tutorial covers:

- discounted utility with log preferences and intertemporal demand;
- comparative statics in the discount factor δ and interest rate r;
- borrowing vs saving with different interest rates (kinked intertemporal budget sets);
- ranking assets by present value and identifying arbitrage;
- no-arbitrage pricing and comparative statics of price ratios.

1 Discounted Utility

Suppose utility is $U(x_0, x_1) = u(x_0) + \delta u(x_1)$ where $u(x) = \ln(x)$. Recall that $u'(x) = \frac{1}{x}$. Suppose the consumer buys consumption to maximize utility subject to a budget constraint

$$p_0 x_0 + p_1 x_1 = m.$$

Thus, when the consumer maximizes utility his demand will satisfy

$$\frac{u'(x_0)}{\delta u'(x_1)} = \frac{p_0}{p_1}.$$

1.1 Log utility demand

Find the demand function: $x(p, m) = (x_0(p, m), x_1(p, m))$. So, $x_k(p, m)$ should tell you how much the consumer demands of good k when price is p and expenditure level is m.

Solution

Method 1: Lagrangian and FOCs

Maximise

$$\ln x_0 + \delta \ln x_1 \quad \text{s.t.} \quad p_0 x_0 + p_1 x_1 = m, \qquad x_0 > 0,\ x_1 > 0.$$

Lagrangian:

$$\mathcal{L} = \ln x_0 + \delta \ln x_1 + \lambda(m - p_0 x_0 - p_1 x_1).$$

FOCs:

$$\frac{1}{x_0} - \lambda p_0 = 0, \qquad \frac{\delta}{x_1} - \lambda p_1 = 0, \qquad p_0 x_0 + p_1 x_1 = m.$$

From the first two,

$$\lambda = \frac{1}{p_0 x_0} = \frac{\delta}{p_1 x_1} \quad \Rightarrow \quad p_1 x_1 = \delta p_0 x_0.$$

Substitute into the budget:

$$m = p_0 x_0 + p_1 x_1 = p_0 x_0 + \delta p_0 x_0 = (1+\delta) p_0 x_0 \Rightarrow x_0(p, m) = \frac{m}{(1+\delta) p_0}.$$

Then

$$x_1(p, m) = \frac{\delta p_0}{p_1} x_0 = \frac{\delta p_0}{p_1} \cdot \frac{m}{(1+\delta) p_0} = \frac{\delta m}{(1+\delta) p_1}.$$

$$\boxed{x_0(p, m) = \frac{m}{(1+\delta) p_0}, \qquad x_1(p, m) = \frac{\delta m}{(1+\delta) p_1}}$$

Method 2: Spending-share (ratio) method

The FOC condition given implies

$$\frac{(1/x_0)}{\delta(1/x_1)} = \frac{p_0}{p_1} \quad \Rightarrow \quad \frac{x_1}{\delta x_0} = \frac{p_0}{p_1} \quad \Rightarrow \quad p_1 x_1 = \delta p_0 x_0.$$

Let $S_0 = p_0 x_0$ and $S_1 = p_1 x_1$. Then $S_1 = \delta S_0$ and $S_0 + S_1 = m$, so

$$(1+\delta)S_0 = m \Rightarrow S_0 = \frac{m}{1+\delta}, \quad S_1 = \frac{\delta m}{1+\delta}.$$

Hence

$$x_0 = \frac{S_0}{p_0} = \frac{m}{(1+\delta)p_0}, \qquad x_1 = \frac{S_1}{p_1} = \frac{\delta m}{(1+\delta)p_1}.$$

$$\boxed{x_0(p,m) = \frac{m}{(1+\delta)p_0}, \qquad x_1(p,m) = \frac{\delta m}{(1+\delta)p_1}}$$

1.2 Effect of δ on x_0

What happens to the demand for good 0 as δ increases?

Solution

Method 1: Differentiate the closed-form demand

From the demand,

$$x_0(\delta) = \frac{m}{(1+\delta)p_0} \quad \Rightarrow \quad \frac{\partial x_0}{\partial \delta} = -\frac{m}{p_0} \cdot \frac{1}{(1+\delta)^2} < 0.$$

$$\boxed{\text{As } \delta \text{ increases, } x_0 \text{ decreases.}}$$

Method 2: Expenditure share argument

The spending ratio satisfies

$$\frac{p_1 x_1}{p_0 x_0} = \delta.$$

As δ rises, the consumer allocates a larger share to good 1 relative to good 0, holding total expenditure m fixed. Since $p_0 x_0 = m/(1+\delta)$ is decreasing in δ, x_0 decreases.

$$\boxed{\text{As } \delta \uparrow, \text{ the consumer shifts spending toward period 1, so } x_0 \downarrow.}$$

1.3 Effect of r with prices $p = (1, \frac{1}{1+r})$

Suppose prices take the form $p = \left(1, \frac{1}{1+r}\right)$. What happens to the demand for good 0 as r increases (equivalently what happens to the demand for good 0 when p_1 decreases)?

Solution

Method 1: Plug into $x_0(p, m)$ with fixed m

The demand is

$$x_0(p, m) = \frac{m}{(1+\delta)p_0}.$$

Here $p_0 = 1$, so

$$x_0 = \frac{m}{1+\delta},$$

which does not depend on $p_1 = \frac{1}{1+r}$ (and hence does not depend on r) when m is fixed.

$$\boxed{\text{Holding } m \text{ fixed, } x_0 \text{ does not change as } r \text{ increases.}}$$

Method 2: Eliminate p_1 using the FOC

From the FOC, $p_1 x_1 = \delta p_0 x_0$. Substitute into the budget:

$$m = p_0 x_0 + p_1 x_1 = p_0 x_0 + \delta p_0 x_0 = (1+\delta)p_0 x_0 \Rightarrow x_0 = \frac{m}{(1+\delta)p_0},$$

which does not involve p_1 at all. Therefore changing r (changing p_1) has no effect on x_0 if m is fixed.

$$\boxed{\text{Holding } m \text{ fixed, } x_0 \text{ is independent of } r.}$$

1.4 Present value of income

Suppose the consumer has income $(y_0, y_1) = (0, 100)$ and the interest rate is r. What is the present value of this income?

Solution

Method 1: Discount the period-1 income

$$PV = y_0 + \frac{y_1}{1+r} = 0 + \frac{100}{1+r}.$$

$$\boxed{PV = \frac{100}{1+r}}$$

Method 2: Replication by saving/borrowing

Find M such that saving M at $t = 0$ yields 100 at $t = 1$:

$$M(1+r) = 100 \Rightarrow M = \frac{100}{1+r}.$$

$$\boxed{PV = \frac{100}{1+r}}$$

1.5 Effect of r when income is $(0, 100)$

Continue to assume the consumer has income $(0, 100)$. What happens to the consumer's demand for good 0 as r increases (note that increasing r changes the present value of the income stream)?

Solution

Method 1: Treat expenditure as present value $m(r)$

With income $(0, 100)$, the present value is

$$m(r) = \frac{100}{1+r}.$$

With prices $p = \left(1, \frac{1}{1+r}\right)$, we have $p_0 = 1$, so

$$x_0(r) = \frac{m(r)}{1+\delta} = \frac{1}{1+\delta} \cdot \frac{100}{1+r}.$$

Since $\frac{100}{1+r}$ decreases in r, $x_0(r)$ decreases in r.

$$\boxed{\text{As } r \text{ increases, } x_0 \text{ decreases.}}$$

Method 2: Monotonicity via PV

The PV of the income stream is strictly decreasing:

$$PV(r) = \frac{100}{1+r} \downarrow \text{ as } r \uparrow .$$

But $x_0 = \frac{m}{1+\delta}$ when $p_0 = 1$. Hence x_0 moves one-for-one with $m = PV(r)$, so x_0 decreases.

$$\boxed{\text{As } r \uparrow,\ PV \downarrow \Rightarrow x_0 \downarrow .}$$

1.6 Numerical demand with given δ and r

Assume the consumer has income $(0, 24)$. Suppose that $\delta = \frac{1}{2}$ and that the interest rate is $r = 1$. How much will be demanded of good 0 and 1?

Solution

Method 1: Compute PV, then apply demand

Present value:

$$m = \frac{24}{1+r} = \frac{24}{2} = 12.$$

Prices: $p_0 = 1$, $p_1 = \frac{1}{1+r} = \frac{1}{2}$. With $\delta = \frac{1}{2}$,

$$x_0 = \frac{m}{(1+\delta)p_0} = \frac{12}{(1+\frac{1}{2})\cdot 1} = \frac{12}{3/2} = 8,$$

$$x_1 = \frac{\delta m}{(1+\delta)p_1} = \frac{\frac{1}{2}\cdot 12}{\frac{3}{2}\cdot\frac{1}{2}} = \frac{6}{3/4} = 8.$$

$$\boxed{x_0 = 8, \qquad x_1 = 8}$$

Method 2: Use FOC ratio and budget directly

FOC gives $p_1 x_1 = \delta p_0 x_0$. Here $\delta = \frac{1}{2}$, $p_0 = 1$, $p_1 = \frac{1}{2}$:

$$\frac{1}{2}x_1 = \frac{1}{2}x_0 \Rightarrow x_1 = x_0.$$

Budget uses PV $m = 12$:

$$x_0 + \frac{1}{2}x_1 = 12.$$

Substitute $x_1 = x_0$:

$$x_0 + \frac{1}{2}x_0 = \frac{3}{2}x_0 = 12 \Rightarrow x_0 = 8 \Rightarrow x_1 = 8.$$

$$\boxed{x_0 = 8, \qquad x_1 = 8}$$

2 Interest rates

Suppose you can borrow at interest rate r_b and save at rate r_s. You are endowed with $y = (1, 1)$.

2.1 Budget line with borrowing expensive

Draw the budget line when $2 = r_b > r_s = 1$ (note that the budget line is not in-fact a straight line).

Solution

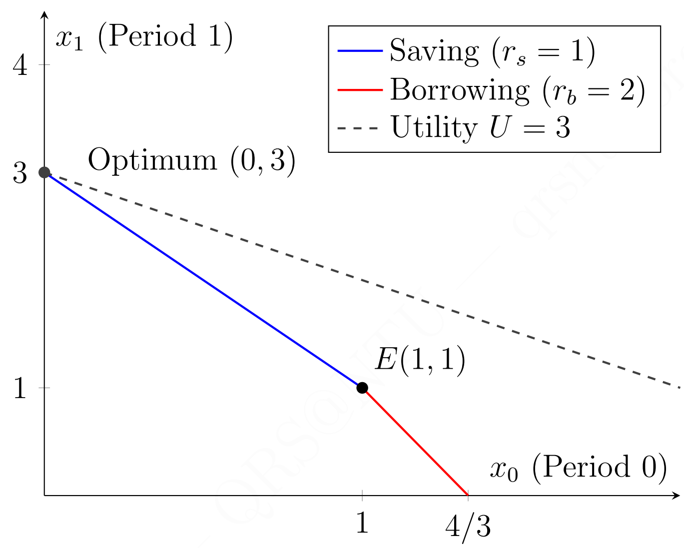

Figure 1: Budget frontier and optimal choice with expensive borrowing ($r_b = 2, r_s = 1$).

Let x_0 be period-0 consumption and x_1 period-1 consumption. Endowment is $(1, 1)$.

Method 1: Derive saving and borrowing pieces

Saving region ($x_0 \le 1$). Saving $s = 1 - x_0 \ge 0$ yields $(1 + r_s)s$ next period:

$$x_1 = 1 + (1 + r_s)(1 - x_0).$$

With $r_s = 1$:

$$x_1 = 1 + 2(1 - x_0) = 3 - 2x_0, \qquad (0 \le x_0 \le 1).$$

Borrowing region ($x_0 \ge 1$). Borrowing $b = x_0 - 1 \ge 0$ requires repayment $(1 + r_b)b$:

$$x_1 = 1 - (1 + r_b)(x_0 - 1).$$

With $r_b = 2$:

$$x_1 = 1 - 3(x_0 - 1) = 4 - 3x_0, \qquad (1 \le x_0 \le 4/3).$$

$$\boxed{\text{Budget frontier: } x_1 = 3 - 2x_0 \ (x_0 \le 1), \quad x_1 = 4 - 3x_0 \ (x_0 \ge 1).}$$

Method 2: Slopes through the endowment point

The frontier passes through $(1, 1)$. Slopes are $-(1+r_s) = -2$ when saving and $-(1+r_b) = -3$ when borrowing. Hence:

$$x_1 - 1 = -2(x_0 - 1) \Rightarrow x_1 = 3 - 2x_0 \ (x_0 \leq 1),$$

$$x_1 - 1 = -3(x_0 - 1) \Rightarrow x_1 = 4 - 3x_0 \ (x_0 \geq 1).$$

$$\boxed{\text{Same piecewise budget frontier.}}$$

2.2 Budget line with saving more attractive

Draw the budget line when $r'_s = 2$ and $r'_b = 1$.

Solution

Method 1: Piecewise derivation

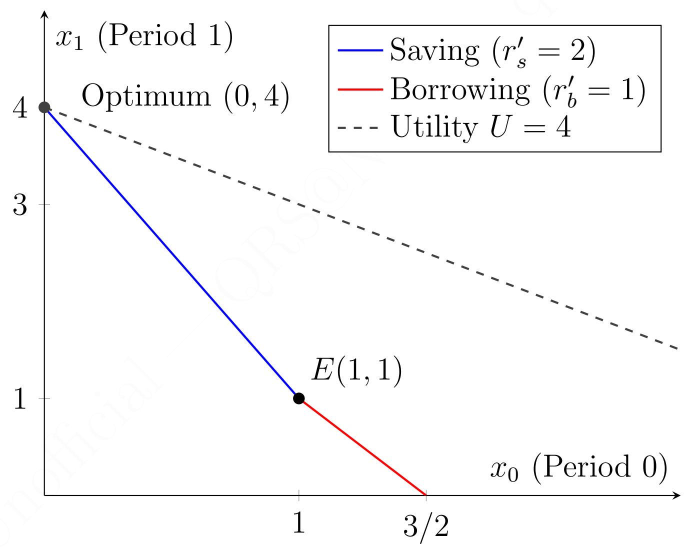

Figure 2: Budget frontier and optimal choice with attractive saving $(r'_b = 1, r'_s = 2)$.

Saving $(x_0 \leq 1)$**.**

$$x_1 = 1 + (1 + r'_s)(1 - x_0) = 1 + 3(1 - x_0) = 4 - 3x_0.$$

Borrowing $(x_0 \geq 1)$**.**

$$x_1 = 1 - (1 + r'_b)(x_0 - 1) = 1 - 2(x_0 - 1) = 3 - 2x_0.$$

$$\boxed{\text{Budget frontier: } x_1 = 4 - 3x_0 \ (x_0 \leq 1), \quad x_1 = 3 - 2x_0 \ (x_0 \geq 1).}$$

Method 2: Slopes through $(1, 1)$

Saving slope is $-(1 + r_s') = -3$ and borrowing slope is $-(1 + r_b') = -2$. Through $(1, 1)$:

$$x_1 - 1 = -3(x_0 - 1) \Rightarrow x_1 = 4 - 3x_0, \quad x_1 - 1 = -2(x_0 - 1) \Rightarrow x_1 = 3 - 2x_0.$$

$$\boxed{\text{Same piecewise budget frontier.}}$$

2.3 Preference over interest-rate regimes with linear utility

Suppose the consumer has $U(x_0, x_1) = x_0 + x_1$ (note discount factor $\delta = 1$). Does the consumer prefer interest rates (r_b, r_s) or (r_b', r_s')? That is, under which pair of interest rates can the consumer buy a bundle with greater utility?

Solution

Method 1: Evaluate utility at key corners

With linear utility and positive coefficients, the optimum occurs at an extreme point.

Regime $(r_b, r_s) = (2, 1)$. Corners: $(0, 3)$, $(1, 1)$, $(4/3, 0)$.

$$U(0, 3) = 3, \quad U(1, 1) = 2, \quad U(4/3, 0) = 4/3.$$

Maximum $= 3$.

Regime $(r_b', r_s') = (1, 2)$. Corners: $(0, 4)$, $(1, 1)$, $(3/2, 0)$.

$$U(0, 4) = 4, \quad U(1, 1) = 2, \quad U(3/2, 0) = 3/2.$$

Maximum $= 4$.

$$\boxed{\text{The consumer prefers } (r_b', r_s') \text{ since } 4 > 3.}$$

Method 2: Maximise along each piece (monotonicity)

For $(2, 1)$: on saving piece $x_1 = 3 - 2x_0$, $U = 3 - x_0$ maximised at $x_0 = 0 \Rightarrow 3$; borrowing piece yields at most 2. For $(1, 2)$: on saving piece $x_1 = 4 - 3x_0$, $U = 4 - 2x_0$ maximised at $x_0 = 0 \Rightarrow 4$. Thus $(1, 2)$ dominates.

$$\boxed{\text{Prefer } (r_b', r_s').}$$

3 Comparing assets

Consider three assets: $y^A = (1,0,0)$, $y^B = (0,3,0)$, and $y^C = (0,0,4)$.

3.1 Best asset by present value

For what interest rates r is A the best? What about B and C? To answer consider the present values.

Solution

Let

$$PV(y;r) = y_0 + \frac{y_1}{1+r} + \frac{y_2}{(1+r)^2}.$$

Then

$$PV_A = 1, \qquad PV_B = \frac{3}{1+r}, \qquad PV_C = \frac{4}{(1+r)^2}.$$

Method 1: Pairwise comparisons

B vs C:

$$\frac{3}{1+r} \geq \frac{4}{(1+r)^2} \iff 3(1+r) \geq 4 \iff r \geq \frac{1}{3}.$$

A vs B:

$$1 \geq \frac{3}{1+r} \iff 1+r \geq 3 \iff r \geq 2.$$

A vs C:

$$1 \geq \frac{4}{(1+r)^2} \iff (1+r)^2 \geq 4 \iff r \geq 1.$$

Thus:

$$\boxed{C \text{ best if } r < \frac{1}{3}; \quad B \text{ best if } \frac{1}{3} \leq r \leq 2; \quad A \text{ best if } r > 2.}$$

Method 2: Cutoff reasoning

Solve equalities:

$$PV_B = PV_C \Rightarrow r = \frac{1}{3}, \qquad PV_A = PV_B \Rightarrow r = 2.$$

As $r \to 0$, $PV_C \to 4$ is largest; as $r \to \infty$, $PV_A = 1$ dominates. Hence the ordering must be C then B then A across the two cutoffs.

$$\boxed{\text{Same cutoff partition: } r = \frac{1}{3},\ 2.}$$

3.2 Effect of endowment $(2, 0, 0)$ on rankings

Answer the same question when you are endowed with $(2, 0, 0)$. That is, if you take asset y^A then you get $(2, 0, 0) + y^A$ and so forth.

Solution

Method 1: Additivity of present value

$$PV((2, 0, 0) + y; r) = PV(2, 0, 0; r) + PV(y; r) = 2 + PV(y; r).$$

Adding the same constant 2 to all options does not change rankings.

$$\boxed{\text{The ranking (and cutoffs) are unchanged.}}$$

Method 2: Pairwise comparisons cancel the endowment

For any two assets $y, \tilde{y}$,

$$PV(e + y; r) \geq PV(e + \tilde{y}; r) \iff PV(y; r) \geq PV(\tilde{y}; r).$$

Thus endowment does not affect which asset is best.

$$\boxed{\text{Same ranking as in Question 3.1.}}$$

3.3 Arbitrage with overpriced asset B

Now, assume that you have no money $(0, 0, 0)$ and asset B costs

$$\frac{4}{1 + r}.$$

Can you find an arbitrage opportunity?

Solution

Method 1: Compare to no-arbitrage PV and construct a strategy

Asset B pays $(0, 3, 0)$, so its no-arbitrage price is

$$PV_B = \frac{3}{1 + r}.$$

Given price $p_B = \frac{4}{1+r} > \frac{3}{1+r}$, B is overpriced.

Arbitrage:

- At $t = 0$: short-sell 1 unit of B, receive $\frac{4}{1+r}$. Invest at rate r.
- At $t = 1$: investment becomes $(1 + r)\frac{4}{1+r} = 4$. Pay the short obligation payoff 3. Keep 1 risk-free.
- No further payoffs.

$$\boxed{\text{Yes: short } B \text{ and invest; sure profit 1 at } t = 1.}$$

3.4 Comparative statics of price ratio p_A/p_C

Let p_A and p_C be the prices of asset A and C in period 0. What happens to the ratio

$$\frac{p_A}{p_C}$$

as r increases (assume p_A and p_C are determined by the no-arbitrage condition)?

Solution

Method 1: Compute and differentiate

No-arbitrage prices:

$$p_A = 1, \qquad p_C = \frac{4}{(1+r)^2}.$$

So

$$\frac{p_A}{p_C} = \frac{(1+r)^2}{4},$$

which is strictly increasing in r.

$$\boxed{\frac{p_A}{p_C} \text{ increases as } r \text{ increases.}}$$

Method 2: Monotonicity argument

As r increases, $(1+r)^2$ increases, so $\frac{(1+r)^2}{4}$ increases. Equivalently, p_C falls with r while p_A is constant, so the ratio rises.

$$\boxed{\text{As } r\uparrow,\ p_C\downarrow \text{ and } p_A = 1, \text{ so } p_A/p_C\uparrow.}$$

Method 3: Partial derivative and rate of increase

From no-arbitrage,

$$\frac{p_A}{p_C} = \frac{(1+r)^2}{4}.$$

Differentiate with respect to r:

$$\frac{\partial}{\partial r}\left(\frac{p_A}{p_C}\right) = \frac{\partial}{\partial r}\left(\frac{(1+r)^2}{4}\right) = \frac{1}{4}\cdot 2(1+r) = \frac{1+r}{2}.$$

Since $1+r>0$ (in particular for $r>-1$, and certainly for standard interest rates $r \geq 0$),

$$\frac{\partial}{\partial r}\left(\frac{p_A}{p_C}\right) = \frac{1+r}{2} > 0,$$

$$\boxed{\frac{p_A}{p_C} \text{ increases at the rate } \frac{1+r}{2} \text{ as } r \text{ increases.}}$$